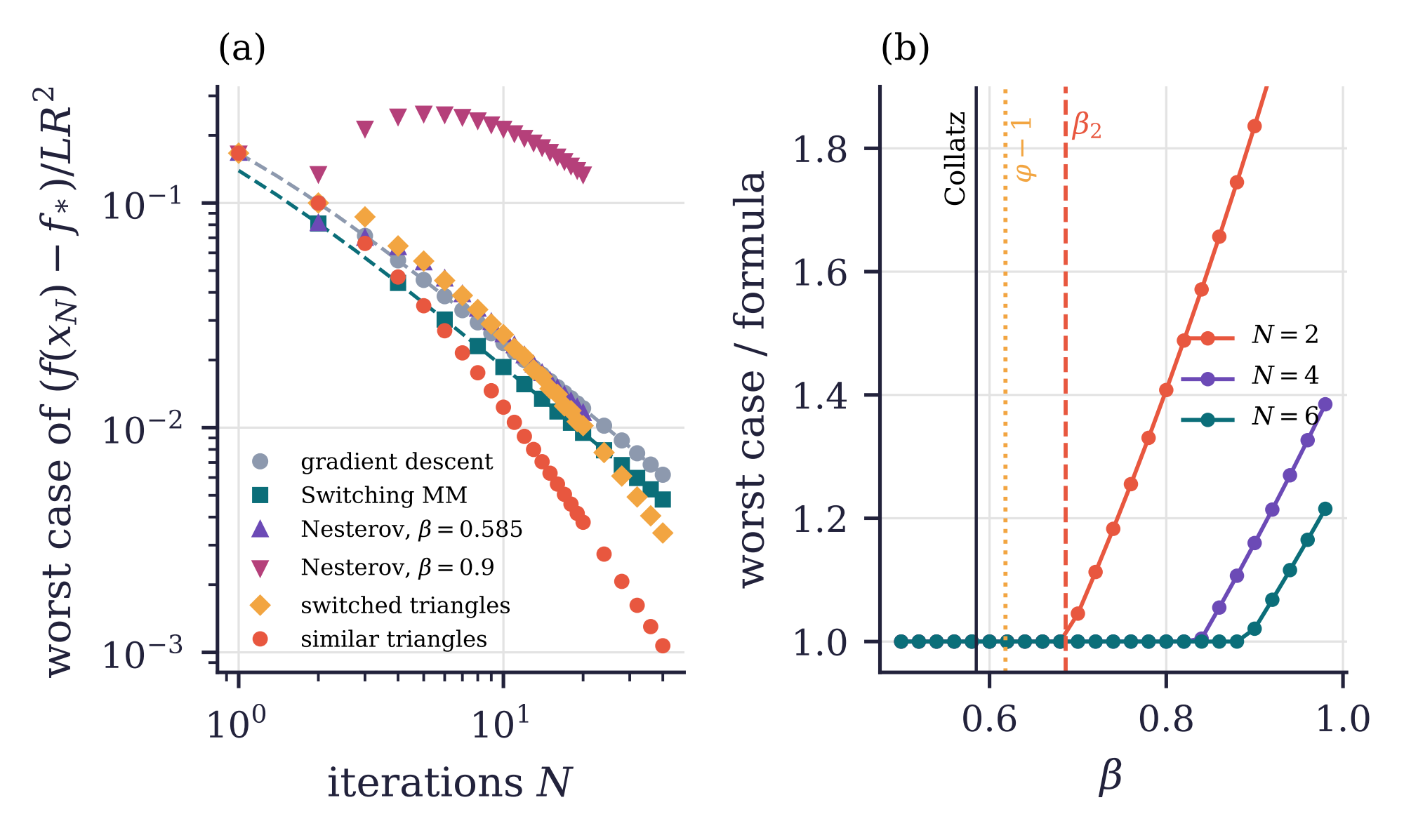
(a)
worst case of $(f(x_N) - f_*)/LR^2$
10^{-1}
10^{-2}
10^{-3}
10^0
10^1
iterations N
gradient descent
Switching MM
Nesterov, $\beta = 0.585$
Nesterov, $\beta = 0.9$
switched triangles
similar triangles
(b)
worst case / formula
1.8
1.6
1.4
1.2
1.0
0.6
0.8
1.0
β
Collatz
$\varphi - 1$
β_2
$N = 2$
$N = 4$
$N = 6$